

Elizabeth "Eli" Margolin

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SUMMARY

PhD candidate at the University of Pennsylvania building zero-knowledge proof systems at the intersection of programming languages and cryptography, with publications at IEEE S&P, USENIX Security, and SOSP. Three years of security engineering at Meta. Current work applies these tools to AI accountability, including zkTLS provenance for LLM responses.

TECHNICAL SKILLS

Cryptography: zero-knowledge proofs; secure two-party and multi-party computation; homomorphic encryption; differential privacy.

Programming: Rust, C++, Python; interactive theorem proving (Rocq, Lean).

Security: threat modeling; abuse and insider-threat detection; security-sensitive code review.

EXPERIENCE

University of Pennsylvania — PhD Researcher, Cryptography & Programming Languages 2021–present

- Designed and built zero-knowledge proof systems for formal languages and zkTLS protocols that let users prove facts about private data and real web sessions; owned the work from problem formulation and protocol design through open-source implementation and evaluation.
- Designed and built a planner that automatically compiles differentially private analytics queries into secure multi-party and homomorphic computations at billion-device scale.
- Work published at IEEE S&P, USENIX Security, and SOSP.
- Advised by Sebastian Angel. Dissertation: *Zero Knowledge Proofs of Formal Languages and Their Applications*.

Brave Software — Research Intern Summer 2024

- Designed zero-knowledge protocols for privacy-preserving fraud detection over authenticated web traffic, taking research designs through to deployable implementations.
- Built and hardened cryptographic tooling to interoperate with modern TLS stacks.

Meta Platforms — Security Engineer 2019–2021 (Contingent Worker, 2022)

- Performed threat modeling and security analysis to design and own company-wide insider-threat and abuse-detection tooling for security and legal investigations teams.
- Scaled alert-processing pipelines through automation and led security and privacy reviews with Legal and Policy, raising throughput while cutting false positives and data footprint.

SELECTED PUBLICATIONS

Coral: Fast Succinct Non-Interactive Zero-Knowledge CFG Proofs *IEEE S&P 2026*
S. Angel, S. Celi, E. Margolin*, P. Mishra, M. Sander, J. Woods

Reef: Fast Succinct Non-Interactive Zero-Knowledge Regex Proofs *USENIX Security 2024*
S. Angel, E. Ioannidis, E. Margolin*, S. Setty, J. Woods

Arboretum: A Planner for Large-Scale Federated Analytics with Differential Privacy *SOSP 2023*
E. Margolin, K. Newatia, E. Roth, T. Luo, A. Haeberlen

* Authors listed in alphabetical order.

EDUCATION

University of Pennsylvania — PhD, Computer and Information Science *In progress*

Duke University — MS, Economics and Computation

Stanford University — BA, Political Science, with Honors in International Security

SELECTED ACTIVITIES

Team USA — National Team Athlete (Rowing); USRowing Athlete Council *2025–present*

- 1st place, 2026 World Rowing Cup III, Lucerne.
- A Finalist, 2025 World Rowing Championships, Shanghai.